

input: a scalar (one number)

input: a vector of n numbers

output: a scalar
(one number)



Case 1: one number (1×1)
 $f'(3) = 6$

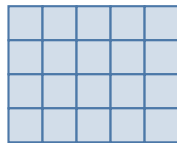


Case 2: row vector ($1 \times n$)
e.g. $n = 2$: $\nabla f(2, 3) = [4, 6]$

output: a vector
of m numbers



Case 3: column vector ($m \times 1$)
e.g. $m = 2$: $\mathbf{g}'(1) = [1, 2]^T$



Case 4: matrix ($m \times n$)
e.g. $m = n = 2$: Jacobian J

each grid: the shape of the derivative, one row per output, one column per input